

Team update — unified-data pivot (Het, 2026-07-23)

Nothing is pushed yet — this is for your review before it lands. Short version of what I changed, why, and what each of you should sanity-check in your phase. Full rationale + numbers: [docs/unified_data_architecture.md](https://docs.unified_data_architecture.md).

Why I changed anything

1. **MIMIC-IV got rejected** (we finished CITI, PhysioNet still denied us) — so real discharge summaries are off the table for this deadline.
2. **The old Kaggle set has no signal.** Every model scored AUROC ≈ 0.5 on it — that's *random*, not "structured features are weak." Our thesis was untestable.
3. **The four phases were on three unrelated datasets**, so the cross-phase ablation (the whole point of the project) wasn't apples-to-apples.

The fix: unify all four phases onto Phase 4's existing synthetic generator, and extend it to emit a **clinical note per claim**. One dataset, one population, so the ablation is finally valid — and it clears both of Prof. Toutiaee's feedback points by construction (one population = no distribution mismatch; owned rule = auditable, noise-swept labels).

What did NOT change: the proposal's scope, the 4 phases, the methods, the eval harness, ~19% prevalence. Only the *data substrate* changed.

What changed, by file

Shared / generator (please skim): - `phase4_rag_agentic/src/data_gen.py` — added a synthetic `clinical_note` per claim + a new latent driver `_necessity_documented`, and a `generate_dataset()` wrapper all phases now call. Structured fields are byte-identical to before (I used a separate RNG for the note text). - `phase4_rag_agentic/src/labeling.py` — added one driver, `undocumented_necessity` (weight 1.6, CO-50/N115), so the note carries a real denial signal. - `shared/utils/eval.py` — added `bootstrap_auroc_lift()` (paired bootstrap CI + p-value on a lift). Purely additive. - `phase4_rag_agentic/src/retriever.py` — silenced a benign numpy warning (`errstate guard`); **numbers unchanged**.

New files (my phases + the runner): - `phase1_baseline/src/train.py` — Phase 1 (LR/DT + SMOTE + RFE + k-fold) ported onto the unified data. - `phase2_gbm_shap/src/train.py` — Phase 2 (GBM/RF + SHAP) ported onto the unified data. - `phase3_clinicalbert/src/{text_encoder, pipeline}.py` — Phase 3 (TF-IDF/ClinicalBERT fused with structured, empty-note ablation, cost, calibration). - `phase3_clinicalbert/src/llm_baseline.py` — the "GPT-4 zero-shot" baseline, on Claude (gated on a key; no-ops without one). - `scripts/run_all_phases.py` — the unified 4-way ablation table + figure. - `docs/unified_data_architecture.md` — full design doc.

■ What each of you should check

- **Sruthilaya (Phase 4):** I edited your `labeling.py` and `data_gen.py`. Your **retrieval lift is preserved** (+0.022, structured 0.733 → augmented 0.755) and your whole pipeline/rigor still runs — but the new driver shifted the absolute numbers a little and raised the oracle ceiling (0.869 → 0.883). Artifacts are refreshed. Please confirm you're OK with the `labeling.py` change (it's your file).
- **Nainica (Phase 2):** I added `phase2_gbm_shap/src/train.py` running your GBM/RF + SHAP on the unified data — it now shows **real signal** (GBM 0.734, RF 0.731) instead of the 0.5 on Kaggle, and SHAP attributes to filing/billed/CPT/payer as expected. Please review it and adjust to match how you'd have written it.

The payoff — unified ablation (one dataset, temporal test)

Model	AUROC
LLM zero-shot (Claude Sonnet 5, note only)	0.606
structured — Phase 1/2	0.733
+ text — Phase 3	0.769 (+0.036, 95% CI [0.026, 0.046], p<0.0005)
+ retrieval — Phase 4	0.755 (+0.022)
+ both	0.795 (+0.063)
oracle ceiling	0.883

Text and retrieval recover **different** latent signals and stack — that's the "codes → context" story, now measurable and statistically clear.

How to run it

```
pip install -e .          # deps are declared per phase
python scripts/run_all_phases.py  # the unified ablation table + figure
```

Figures/metrics land in artifacts/ and each phase*/artifacts/. Happy to walk through any of it or change anything before we push — flag it here.